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RESEARCH ARTICLE
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The Impacts of Micro-Porosity and Mineralogical Texture on Fractured Rock Alteration

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Key Points:

- Spatial variations in altered layers affect the evolution of bulk reaction and permeability in fractured media
- The characteristic length of reactive mineral clusters dominates the internal texture of the altered layer and permeability enhancement
- The presence of micro-porosity in the matrix promotes mineral accessibility, reactive transport, and thus thickening of altered layers

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Geochemically driven alterations of fractures in multi-mineral media can create altered layers (ALs) at the fracture-matrix interface. Spatial variations in the AL significantly influence mass transfer across the interface, and the hydraulic and mechanical properties of the fractured medium. A real-rock based microfluidic experiment reported spatial variations in AL thickness despite the initially smooth fracture surface, suggesting potential effects of matrix heterogeneity on AL development. However, the respective contribution of structural and mineralogical characteristics is still poorly understood. Using the microfluidic experimental data and a micro-continuum reactive transport model, we systematically evaluated how micro-porosity and initial mineral texture impact AL development and thus the overall reactive transport behaviors. Our simulation results confirmed that the extent of AL spatial variations, mainly controlled by mineralogical texture, influences the evolution of reaction and permeability in different ways. Accounting for spatial heterogeneity in mineral distribution produces "channeling" structures in ALs and lower overall reaction (by up to 35.6%), but larger permeability increase (by up to 9.8%). The characteristic length of the reactive mineral cluster was observed to dominate the internal texture of ALs. Whereas the presence of micro-porosity can enhance mineral accessibility via improving connectivity for flow and transport, and lead to both higher bulk reaction, that is, thicker ALs, and permeability enhancement. Considerations of surface roughness with characteristic length on the same order of magnitude as mineral texture did not change the overall development of AL, which further highlights the importance of accounting for rock matrix properties in predicting long-term evolution of fractured media. The resulting spatial variations of ALs and their impacts on bulk properties, however, are expected to be further complicated by the coupling of chemical and mechanical processes, and may trigger matrix disaggregation, erosion and other mechanisms of fractured media alteration.

1. Introduction

Fracture morphology and hydrodynamic properties control fluid migration and solute transport in the Earth subsurface, and can be altered by water-rock interactions (Deng & Spycher, 2019; Ellis et al., 2013; Renard, 2021). In energy and environmental systems such as geologic carbon storage, enhanced oil recovery, and geothermal extraction, the changes in physicochemical conditions may trigger far-from-equilibrium reactions at the fracture-matrix interface (Beckingham, 2017; Molins et al., 2014). The rate and the extent of the subsequent alteration are controlled by the interplay between local fluid dynamics, matrix diffusion, fluid chemistry, mineralogy, etc (Deng & Spycher, 2019; Kang et al., 2014; Steefel et al., 2015). Therefore, fundamental understanding of coupled processes in fractured rocks is essential for improving our predictive capability for assessing engineered and environmental systems (Carroll et al., 2016; Deng et al., 2016).

Previous studies have demonstrated that geometrical changes in fractures induced by mineral dissolution in the surrounding matrix depend on the relative rates of advection, diffusion and reaction (Liu et al., 2017; Szymczak & Ladd, 2009). Typically, as flow rate increases, fracture alteration shifts from compact dissolution to wormhole formation to uniform dissolution (Brunet et al., 2016). But heterogeneity in pore structure and mineral composition of the bordering matrix may lead to different alteration characteristics compared to homogeneous porous media under similar conditions (Al-Khulaifi et al., 2019; Andrews & Navarre-Sitchler, 2021; Chen et al., 2014). In single mineral systems, increased initial pore structural heterogeneity (e.g., with pore sizes spanning over several length scales) produces unstable dissolution fronts, a reduced overall reaction, and a larger exponent for

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the porosity-permeability power law, especially at low flow rates (Al-Khulaifi et al., 2018; Jung & Navarre-Sitchler, 2018).

In multi-mineral systems, spatial distributions of the reactive minerals can create complex fracture morphologies (Jones & Detwiler, 2019; Min et al., 2016). For banded mineral distributions, preferential dissolution of the fast-reacting minerals can result in a comb-tooth structure that affects the subsequent reaction through surface area, hydrodynamics and transport (Deng, Molins, et al., 2018; Deng, Steefel, et al., 2018; Fazeli et al., 2018). In another example, when minerals of differing reactivities are mixed, an altered layer (AL) adjacent to the fracture surface develops following dissolution of the fast-reacting minerals. The remaining slow-/non-reacting minerals serve as a diffusion barrier, resulting in an increasingly reduced reaction rate, and thus mitigating fracture permeability enhancement (Deng et al., 2017; Garcia-Rios et al., 2017; Noiriel et al., 2013). In particular, the porosity-permeability relationship may not follow a power law relation and the permeability even approaches a constant value at the later stage of reaction (Fazeli et al., 2019). Moreover, ALs with a large porosity are structurally weakened, and thus may collapse and detach from the fracture surface (Deng, Molins, et al., 2018; Deng, Steefel, et al., 2018; Liu et al., 2020; Spokas et al., 2019). In these cases, while local apertures increase, the mobilized particles can be transported downstream and accumulate at narrow flow paths, resulting in fracture clogging (Ellis et al., 2013; Noiriel et al., 2007; Spokas et al., 2018). Therefore, the evolution of fracture morphology and permeability is highly dependent on the texture of reactive minerals.

Experiments via state-of-the-art imaging methods, for example, X-ray computed tomography (XCT) and focused ion beam scanning electron microscopy (FIB-SEM), have enabled direct observations of local heterogeneities and resulting geochemical alterations, including the development of the AL (Al-Khulaifi et al., 2017; Singh et al., 2018; Welch et al., 2017). Using an in-operando synchrotron-based microfluidic system, Deng et al. (2020) examined acid erosion of a carbonate rich shale with detailed characterization of the fracture-matrix interface. While the fracture surface was initially flat, the developed AL shows evident spatial variations in thickness, illustrating channeling of the reaction front within the bordering matrix. It was hypothesized that matrix heterogeneity has caused the spatial variations in AL development, further investigations are needed to test the hypothesis and reveal the respective contribution of heterogeneity in pore structure and mineralogy on AL development in order to formulate predictive models at larger scales. This is also the main goal of our work.

Recent development of image-based modeling opens up opportunities to complement experimental observations and fill research gaps regarding AL development (Aljasmi & Sahimi, 2020; Fazeli et al., 2019; Matouš et al., 2017; Noiriel & Soulaine, 2021). However, challenges of the trade-off between image resolution, field-of-view, and computational capability remain, which requires improved co-design of experiments and modeling (Carrillo et al., 2022; Chung et al., 2021). One example is sub-resolution structural and mineralogical features (Ma et al., 2017). In practice, small pores may not be properly resolved due to limitations of image acquisition and processing (Kang et al., 2019; Yang et al., 2022), and they are typically represented by a variable micro-porosity instead. The unresolved pore spaces can largely enhance pore connectivity and thus have a significant impact on fluid flow. For example, permeability can increase by a factor >2 when a $\sim 2\%$ micro-porosity is included (Soulaine et al., 2016). Micro-porosity is a manifestation of the multiscale nature of rock heterogeneity, and its effect on reactive transport processes and the resulting alteration of fractured media is scarcely addressed.

Because of the multiscale nature of structural and mineral heterogeneity in fractures and the bordering matrix, modeling AL development requires the use of a hybrid approach (Jung & Navarre-Sitchler, 2018; Molins & Knabner, 2019). One option is to couple a pore-scale domain for the fracture and apparent pores (i.e., resolved pore space) and a micro-continuum domain for sub-resolution regions (Molins et al., 2019), and a popular implementation of this micro-continuum approach (Soulaine & Tchelepi, 2016; Steefel et al., 2015) is based on the Darcy-Brinkman-Stokes (DBS) equation (Molins & Knabner, 2019; Noiriel & Soulaine, 2021; Soulaine & Tchelepi, 2016; Zhang et al., 2022). Our previous study has used this approach to successfully model reactive transport processes within an established AL for a single-time snapshot (Zhang et al., 2022). The potential of this approach in capturing the temporal evolution of AL with considerations of multi-scale features in pore structure and mineralogy is yet to be evaluated.

In this study, we aim to explore the respective contribution of heterogeneities in pore structures and mineralogy to the development of AL in a flat fracture, and the resulting changes in chemical-physical properties. Based on data from a previous microfluidic experiment, we performed micro-continuum reactive transport modeling to examine dissolution in a naturally heterogeneous rock containing multiple pore sizes and minerals distributed

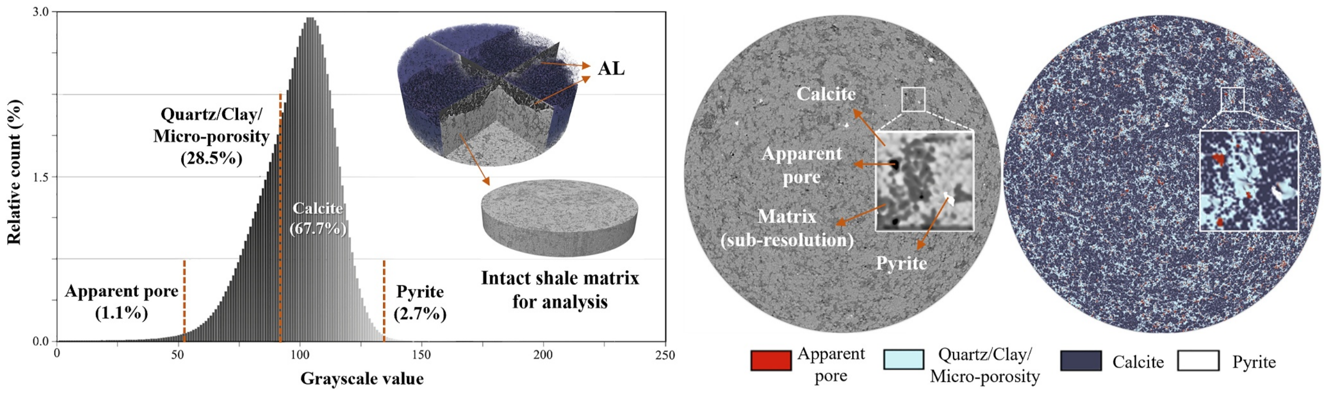

Figure 1. Grayscale histogram of the intact shale matrix, and illustration of the multi-thresholding segmentation of the 3D XCT images for characterization of apparent pores, micro-porosity and mineral phases. The total volume fraction of quartz, clay and micro-porosity is 28.5%.

nonuniformly. The validated model was used to provide a full description of advection, diffusion, and reactions, as well as to enable a flexible representation of multi-scale features. Simulation scenarios with different levels of initial structural and mineralogical heterogeneity were compared to isolate their impacts on AL development. The simulation results also provide insights regarding the level of heterogeneity to be considered for accurate prediction of fractured media evolution.

2. Methods

This modeling investigation builds upon a well-characterized experimental data set (Deng et al., 2020), and here we only provide a brief summary of the experiment. This section also includes information of additional XCT image processing for resolving micro-porosity, and of the model and simulation setup.

2.1. Microfluidic Experiment Data Recap and Further Analyses

The microfluidic cell includes a parallel plate channel carved within the polydimethylsiloxane (PDMS) layer that is bonded to a smooth wafer of the Eagle Ford shale. A hydrochloric acid solution at pH of 2.2 was injected through the channel at 0.2 mL/min (i.e., a linear average velocity of $\sim 7 \times 10^{-4}$ m/s) for 48 hr. This mimics a fracture-matrix interface exposed to a homogenized flow field. Real-time measurements of X-ray transmission through the channel and the rock were collected every 15 min to monitor fracture surface erosion and AL development. Two-dimensional (2D) maps of AL thickness with a spatial resolution of 1.3 μm were derived. Powder X-ray diffraction (PXRD) results confirmed that calcite (70 wt%) is the dominant reactive mineral, and the remaining 30 wt% are quartz, pyrite, and clay minerals. The postmortem XCT images of a sub-plug close to the inlet with a voxel size of 4.6 μm provided detailed morphological characterization of the intact and reacted regions of the rock sample. More information of the experiment was documented in Deng et al. (2020) and its Supporting Information (SI) (Deng et al., 2020).

Here, in order to capture sub-resolution pores, additional XCT imaging analyses were performed on a subsection of the unreacted sample. The gray-level histogram showing no well-separated peaks indicates the existence of mixtures of pore space and solid phases within a single voxel, that is, sub-resolution pores. The Eagle Ford shale has been reported to have pores with diameters of 1–100 nm mainly in clay minerals and an average total porosity of 5% (Cho et al., 2016; Letham & Bustin, 2016; Ojha et al., 2017). Using a multi-thresholding approach, the dark voxels and bright voxels were classified as "apparent pore" and "apparent mineral," respectively; and voxels with intermediate grayscale values were classified as "porous media with micro-porosity" (Figure 1). We assumed that micro-porosity only exists in intermediate grayscale voxels for which the minerals are quartz and clays, and thus the thresholds were chosen such that the resulting proportions of "apparent mineral" (calcite and pyrite) and the "porous media" (quartz and clays, which were not explicitly distinguished due to their similar densities and effective atomic numbers) are consistent with the PXRD mineralogy data. The micro-porosity was assumed to be uniformly distributed in the porous domain, and its value was chosen such that the total porosity is 5%. It should be noted that micro-porosity will be referred to as structural heterogeneity, because itself is a result of pore-size heterogeneity and it is only considered in non-reacting minerals. 2D slices of the segmented data were then used for the simulations.

2.2. Micro-Continuum Reactive Transport Model and Simulation Setup

In the micro-continuum model, the volume fraction of fluid (ε_f) is assigned to each grid cell, and is 1, 0, or a fraction value for the open space (i.e., fracture and macro-pores), minerals, or porous regions, respectively (Bousquet-Melou et al., 2002; Brinkman, 1947). The Darcy Brinkman Stokes equation was solved in COMSOL for the flow field, which was passed to CrunchTope to solve the advection-diffusion-reaction equation and update the ε_f field. Matlab LiveLink was used as the application interface to control the iteration between COMSOL and CrunchTope. A full description of the model and numerical implementation is available in Zhang et al. (2022). In the previous application, the simulations were performed for a time snapshot with no update of the porous media. Here, we enabled dynamic simulations that update fluid chemistry, porosity and permeability in each iteration loop between the flow and reactive transport steps. We also note that the simulation framework has been validated to simulate a benchmark problem with evolving geometry, which shows good agreement with other pore-scale models. Details of the validation can be found in Text S2 of the Supporting Information S1 (Molins et al., 2017, 2021).

In the simulations, we treated minerals other than calcite as non-reactive. Calcite dissolution rate (mol/m^{2}s) was described as follows (Chou et al., 1989):

$$r_{\mathrm{CaCO_3}} = \left(k_1\gamma_{H^+}C_{H^+} + k_2\gamma_{\mathrm{H_2CO_3}^*}C_{\mathrm{H_2CO_3}^*} + k_3\right)\left(1 - \frac{Q_{\mathrm{CaCO_3}}}{K_{\mathrm{CaCO_3}}}\right), \tag{1}$$

where $K_{\mathrm{CaCO_3}}$ is the solubility, $Q_{\mathrm{CaCO_3}}$ is the ion activity products. k_1, k_2, k_3 are the far from equilibrium dissolution rate constants and are 0.89, 5.01×10^{-4}, and 6.6×10^{-7} mol/m^2/s at 25°C, respectively. Reactive surface area (A) of calcite is updated as a function of ε_f and the volume fraction of calcite ($V_{Calcite}$):

$$A = A_0\left[\left(\frac{\varepsilon_f}{\varepsilon_{f,0}}\right)\left(\frac{V_{Calcite}}{V_{Calcite,0}}\right)\right]^{2/3}, \tag{2}$$

where the subscript 0 denotes the initial value, and A_0 is 3,000 m^2/m^3 (Deng et al., 2020). ε_f is updated based on calcite dissolution as follows:

$$\varepsilon_f = 1 - V_{Calcite} - V_{NR}, \tag{3}$$

where V_{NR} is the volume fraction of nonreactive minerals.

Local permeability K (m^2) in the porous grid cells was updated following the Kozeny-Carman (K-C) permeability-porosity relationship (Parkhurst & Appelo, 1999):

$$K^{-1} = K_{const}^{-1}\frac{\left(1-\varepsilon_f\right)^2}{\varepsilon_f{}^3}, \tag{4}$$

To ensure no-slip boundary conditions at solid/fluid interfaces, the input parameter K_{const} of 1×10^{-14} m^2 was chosen such that K within the mineral is at least four orders of magnitude lower than what is in the fluid zone (Soulaine & Tchelepi, 2016). The effective diffusion coefficient (D_{eff}) in the porous region is calculated as:

$$D_{eff} = \varepsilon_f D_0, \tag{5}$$

where D_0 is the diffusion coefficient in water and is 1×10^{-9} m^2/s. The coupling timestep of reactive transport simulation was 0.05 hr, and ε_f, $V_{Calcite}$, K, A, D_{eff} were updated for each coupling time step.

In this work, we focused on a 2D slice from the XCT images using initial conditions, boundary conditions and initial parameters consistent with the experiment. Simulations based on a few additional 2D slices were also

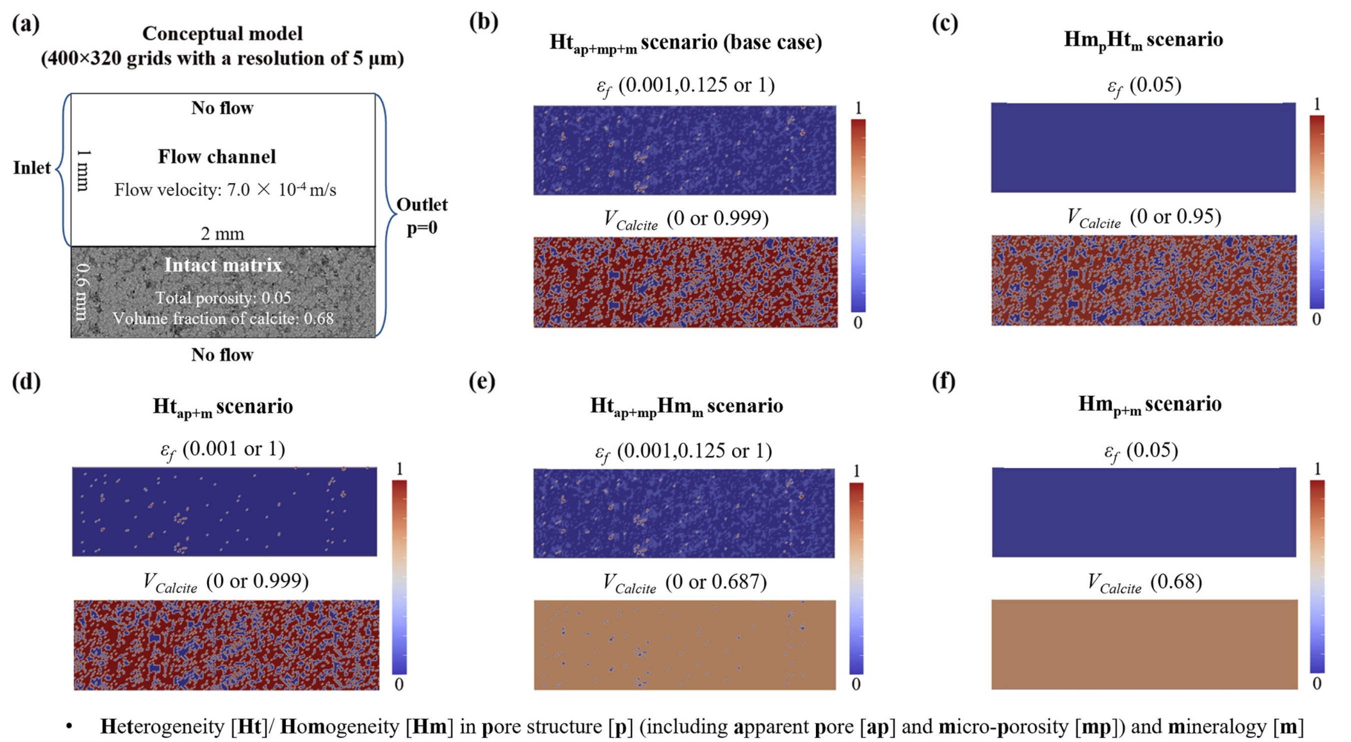

Figure 2. Conceptual illustration of (a) initial and boundary conditions, and the setup of pore structure and mineralogy in intact matrix for different simulation scenarios, that is, (b) the $Ht_{ap+mp+m}$ scenario (base case) following the structural and mineralogical heterogeneity from the CT image, (c) the Hm_pHt_m scenario with spatial heterogeneity only in mineralogy and the porosity uniformly distributed, (d) the Ht_{ap+m} scenario with spatial heterogeneity in mineralogy and apparent pores while ignoring micro-porosity, (e) the $Ht_{ap+mp}Hm_m$ scenario considering homogeneous mineralogy and spatial heterogeneity in apparent pores with micro-porosity, and (f) the Hm_{p+m} scenario considering homogeneous mineralogy and pore structure.

performed to confirm the reliability and representativeness of the modeling results and were detailed in Supporting Information S1. The computational domain includes a 1 mm wide flat fracture and a 0.6 mm thick rock zone. The grid size is 5 μm, comparable to the XCT image resolution (Figure 2). The base case ($Ht_{ap+mp+m}$ scenario) was assigned properties from the segmented images described in Section 2.1, considering heterogeneity in pore structure (p) (including apparent pores (ap) and micro-porosity (mp)) and in mineralogy (m). For the 2D slice used, the volume fractions of apparent pores and calcite are 1.2% and 68.0%, respectively, and the micro-porosity within the porous regions is 12.5%. Results from the base case are compared with the experimental observations, including the bulk reaction rate, average AL thickness, and spatial patterns of the AL as analyzed by semi-variograms. Simulation scenarios with different levels of rock heterogeneity in initial pore structure and mineralogy were formulated for comparison, by considering uniform or non-uniform distributions of apparent pores, micro-porosity, and calcite, to evaluate the impacts of different factors on AL development and thus alteration of the fractured media separately. The initial porosity and volume fraction of calcite in rock matrix across scenarios are also reported in Figures 2b–2f. For cases considering structural spatial heterogeneity, initial ε_f was set as 0.001, 0.125, or 1 in grid cells of mineral phases (dark blue color), micro-porosity (light blue color) and apparent pores (red color), respectively. For cases with spatial mineralogical heterogeneity, $V_{Calcite}$ was ~0.99 (grid cells with reddish colors) (e.g., Ht_{ap+m} scenario), whereas it was ~0.68 when assuming different minerals were fully mixed (e.g., Hm_{p+m} scenario). All the scenarios have the same total porosity (5%) and a total calcite content of 0.68, except for the Ht_{ap+m} case that has a total porosity of 1.2%. Simulation setups for different scenarios are also summarized in Table S1 of the Supporting Information S1.

AL thickness was calculated to compare the rate and extent of AL development across scenarios and with the experiment. For a specific position (x) along the flow direction, the AL thickness (T) at a given time (t) can be calculated as:

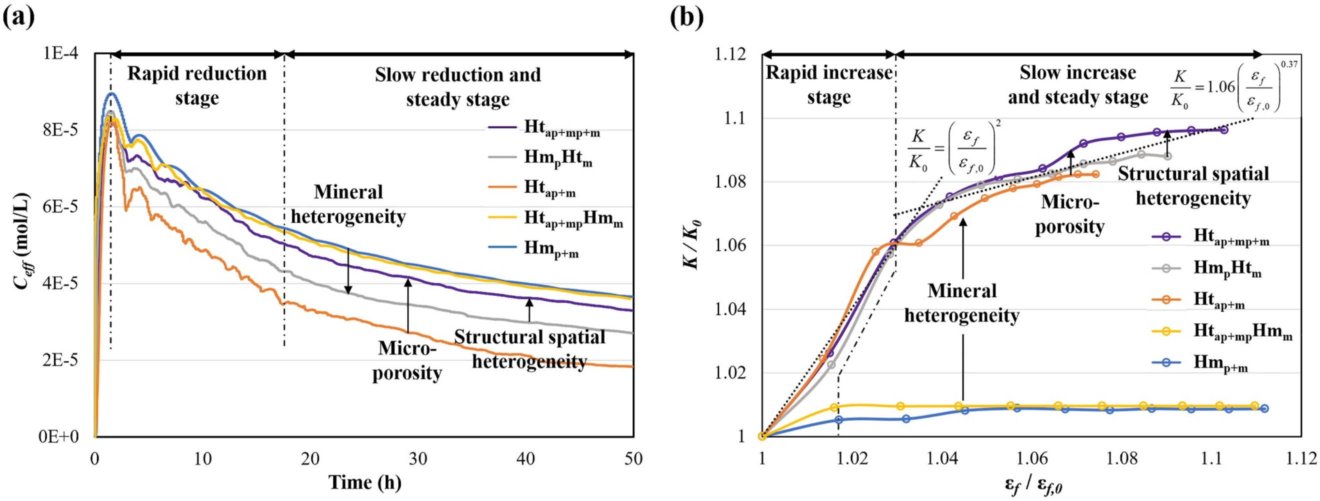

Figure 3. Comparison of (a) the effluent Ca concentration (C_{eff}) and (b) porosity-permeability relationship for different simulation scenarios. In figure (b), the data points are porosity and permeability values calculated for a 5 hr interval, and the dotted lines represent the fitted power law relations. The arrows in this figure help isolate the respective contribution of different structural and mineralogical characteristics.

$$T(x,t) = \frac{\int_{y=0}^{y=Y} \left[\varepsilon_f(x,y,t) - \varepsilon_f(x,y,0) \right] dy\, dx}{\int_{y=0}^{y=Y} V_{Calcite}(x,y,0)\, dx\, dy / Y}, \tag{6}$$

where y is the distance to the fracture surface, and Y is the thickness of intact matrix.

3. Results and Discussion

3.1. Evolution of Physicochemical Properties Induced by AL Development

The effluent Ca concentration (C_{eff}) is tracked to quantify overall reaction rate and its changes along with AL development. The C_{eff} increases first as the acidic fluid displaces the initial DI water, and then decreases because of the well-known diffusion limitation caused by the AL (shown in Figure 3a). For the base case, the effective reaction rate reduces from 3.8×10^{-4} to 2.5×10^{-4} mol/m^2 s in the first 20 hr, which is consistent with the experiment (Deng et al., 2020). This confirms that the numerical model can reproduce the dynamics of bulk behaviors induced by AL development. Similarly, the AL thickens at a decreasing rate. The average AL thickness (calculated from Equation 6) in the base scenario increases from 110 μm (t = 10 hr) to 240 μm (t = 30 hr) to 335 μm (t = 50 hr). The simulated AL thicknesses are comparable to the experimental data, for which the average values increased from 146 μm (t = 10 hr) to 369 μm (t = 30 hr) to 403 μm (t = 40 hr). The slight underestimation of AL thickness may be attributable to the missing dimension in 2D simulations. Comparing the evolution of effective reaction rates and average AL thicknesses between the base simulation and the experiment confirms the predictive capability of this modeling framework for overall behavior.

The extent of the decrease in the overall reaction rate over time varies across scenarios (also summarized in Table S2 of the Supporting Information S1). Uniform distribution of calcite, as in the Hm_{p+m} and $Ht_{ap+mp}Hm_m$ scenarios, resulted in higher reaction rates, with Ca concentrations at hour 50 being higher than those of the Hm_pHt_m case by ∼35.6%. For a given abundance of minerals and the same flow and transport conditions, a more homogeneous distribution of calcite means better connectivity and thus accessibility of the reactive mineral in general, and thus faster overall dissolution reaction. In contrast, removing structural heterogeneity reduces the overall reaction rate, and such impact is more prominent with a heterogeneous mineralogy. When spatial variations in pore structure is ignored (Hm_pHt_m) or simplified by ignoring the micro-porosity (Ht_{ap+m}), the bulk reaction rate at hour 50 is lower than that of the base case by 17.9% and 44.4%, respectively (highlighted in Figure 3a). It further demonstrates that the contribution of micro-porosity to the overall pore connectivity should be accounted for accurate depiction of mineral accessibility and thus reactive transport processes.

In agreement with the bulk reaction rate, the AL thickness by the end of the simulation is higher for the cases with homogeneous calcite distribution (353 and 347 μm for the scenario of the Hm_{p+m} and $Ht_{ap+mp}Hm_{m}$, respectively), and is the lowest with a heterogeneous calcite distribution and without considering micro-porosity (245 μm for the Ht_{ap+m} case). However, the higher reaction rates do not always correspond to larger permeability enhancement. The increase in permeability is more pronounced for scenarios with mineralogical spatial heterogeneity, even though the increase in total porosity is less than that of the scenarios with fully mixed minerals (shown in Figure 3b). Overall, the changes in permeability with respect to changes in total porosity across all scenarios are much more limited compared to what was reported in the literature because permeability of the computational domain evaluated in our study remains to be primarily controlled by the initial fracture opening, but the differences across these scenarios are notable.

Porosity and permeability changes are usually described by a power law relationship, $\frac{K}{K_0} = c\left(\frac{\varepsilon_f}{\varepsilon_{f,0}}\right)^a$, where the exponent a is often assumed to be 3 in fractured domains and the constant c is 1 when the initial fracture is smooth (Al-Khulaifi et al., 2019; Hommel et al., 2018). In our study, the porosity-permeability relationships deviate from the cubic relationship because of the presence of multiple minerals and the AL, and the potential permeability enhancement is dependent on the degrees of heterogeneity. For the scenarios with mineralogical heterogeneity, the initial changes in permeability and porosity for the fractured medium can be fitted by a power law relation with $a = 2$. The increase in permeability levels off with the exponent being reduced to ∼0.37 (shown in Figure 3b) as fracture aperture enlargement is limited by the undissolved minerals in the AL. For the scenarios with homogeneous mineralogy, the fracture permeability reaches a plateau (i.e., $a = 0$) even though the porosity continues to increase, resulting in a difference in permeability enhancement by ∼9.8%. Similarly, not accounting for the complexity in pore structure (e.g., in the Ht_{ap+m} and $Hm_{p}Ht_{m}$ scenarios) can lead to a more limited enhancement in permeability than the corresponding case with the same mineralogical setup, which is particularly evident when the micro-porosity is not considered (i.e., Ht_{ap+m}).

Our results are consistent with previous studies considering only fractures or porous media, which reported that the exponent of the permeability-porosity relationship varies in a broad range of 0.29–28.3 (Fazeli et al., 2018; Menke et al., 2016), and is extremely low when containing nonreactive minerals. For different initial mineralogical and structural characteristics, that is, different 2D slices, the evolution of bulk reaction and permeability show similar trajectories (Figures S4c and S4d in Supporting Information S1).

3.2. Structural and Mineralogical Heterogeneity Controls on AL Development

Our simulation results showed that structural and mineralogical heterogeneity have opposite effects on the overall reaction, while both tend to elevate permeability enhancement. This is related to the interplay between flow, diffusion and reaction, which is manifested largely in AL development. In order to highlight local perturbations in reactive transport arising from matrix spatial heterogeneity, the spatial profiles of AL development are plotted in Figure 4a. The spatial variations of AL thickness increase over time and differ across scenarios, confirming a significant role of matrix heterogeneity in AL development. It is worth noting that varying the degrees of rock heterogeneity can lead to a more pronounced difference in AL thickness distributions compared to that across the slices used in the simulations (shown in Figure S5 of the Supporting Information S1). In order to isolate the impacts of structural and mineralogical heterogeneity on AL spatial features, a comparison of local porosity increase at 50 hr for all simulation scenarios is shown in Figure 4b. In particular, the white dotted lines are where calcite dissolution is the fastest and thus provides an approximation of the reaction front, above which calcite is almost completely depleted. This is typically used to locate the boundary of the AL on XCT images, as the depleted zone can be easily identified. The white solid lines are contours of a very low reaction rate of 1×10^{-8} mol/m^2/s, and serve as an indicator of the exterior boundary of calcite dissolution, that is, where the diffusion front reaches.

3.2.1. Contribution of Structural Heterogeneity

Comparisons of the $Ht_{ap+mp+m}$ (base case), $Hm_{p}Ht_{m}$ and Ht_{ap+m} scenarios provide a direct measure of structural heterogeneity impacts on AL development. The limited propagation and the jaggedness of the reaction front (shown by the dotted line) are most evident when micro-porosity is not considered (i.e., in the Ht_{ap+m} case). These behaviors imply that the slower and the more localized reaction can result from neglecting the contribution of

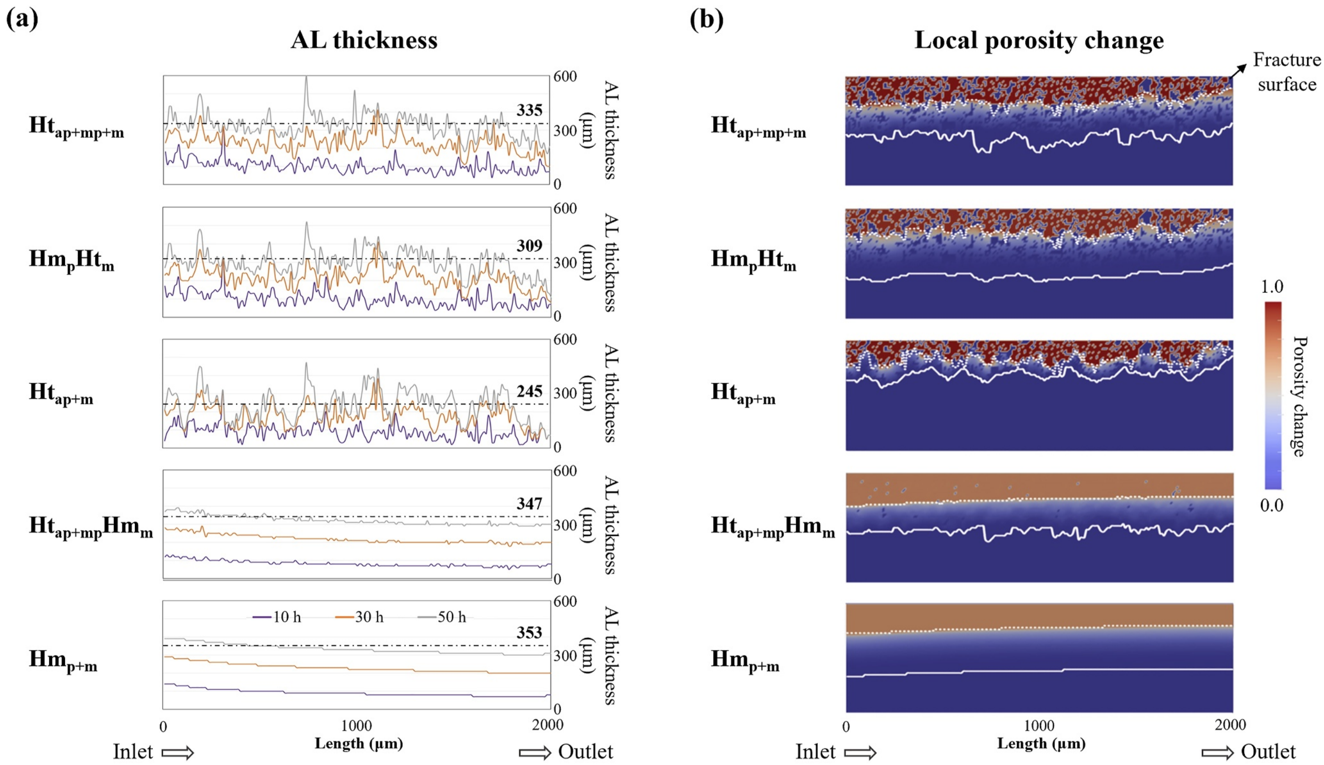

Figure 4. (a) AL thicknesses along the flow direction at 10, 30, 50 hr, respectively, and (b) maps of porosity increase at 50 hr for different scenarios. The dash-dotted lines in (a) indicate the average AL thickness at 50 hr for scenarios. The white dotted lines and white solid lines in (b) show the positions of the maximum reaction rate and a low reaction rate (1×10^{-8} mol/m^2/s), respectively. The zone between two lines is where reaction actively happens at 50 hr.

micro-porosity to pore connectivity, which can also amplify the limitation of mineralogical heterogeneity on reactive transport.

Examination of the structure of calcite-depleted regions, that is, the AL, in these scenarios shows more "channeling" of fast reaction paths, the development of which can more effectively increase permeability of the computational domain. In particular, the largest permeability enhancement can be observed in the base case that resolves both apparent pores and micro-porosity. However, the "channeling" may restrict access to available calcite minerals and thus the overall reaction, even though advective transport may be improved in these regions. Experimental investigation has also confirmed that a considerable decrease in effective reaction rates can result from channel development or wormholing in porous media (Al-Khulaifi et al., 2019).

In addition, the development of the diffusion front (solid lines) is an important indicator of the overall transport as well as the accessibility of reactive minerals. The presence of micro-porosity promotes the migration of the diffusion front, and the non-uniform distribution of micro-porosity in the solid phase triggers the jaggedness in the front shape. The regions between the reaction front and the diffusion front are where calcite is actively dissolving but at a lower rate because of the diffusion limitation. Compared to the base case, the average thickness of the transition zone is 39.5% smaller without considering micro-porosity. The transition zone exhibits a more gradual porosity change for the scenarios with resolved mineralogical heterogeneity (Figure S3 in Supporting Information S1). Since this transition zone would affect the calculation of AL thickness based on the equation described in Section 2.2, which is a common practice when transmission data were used for calculating AL thickness, our observations above demonstrate that the calculation procedure may need to be revisited based on matrix characteristics.

In summary, spatial heterogeneities in pore structure control pore connectivity and thus have a profound effect on the accessibility of reactive minerals. Accounting for micro-porosity is necessary for accurate prediction of both local alteration and overall evolution.

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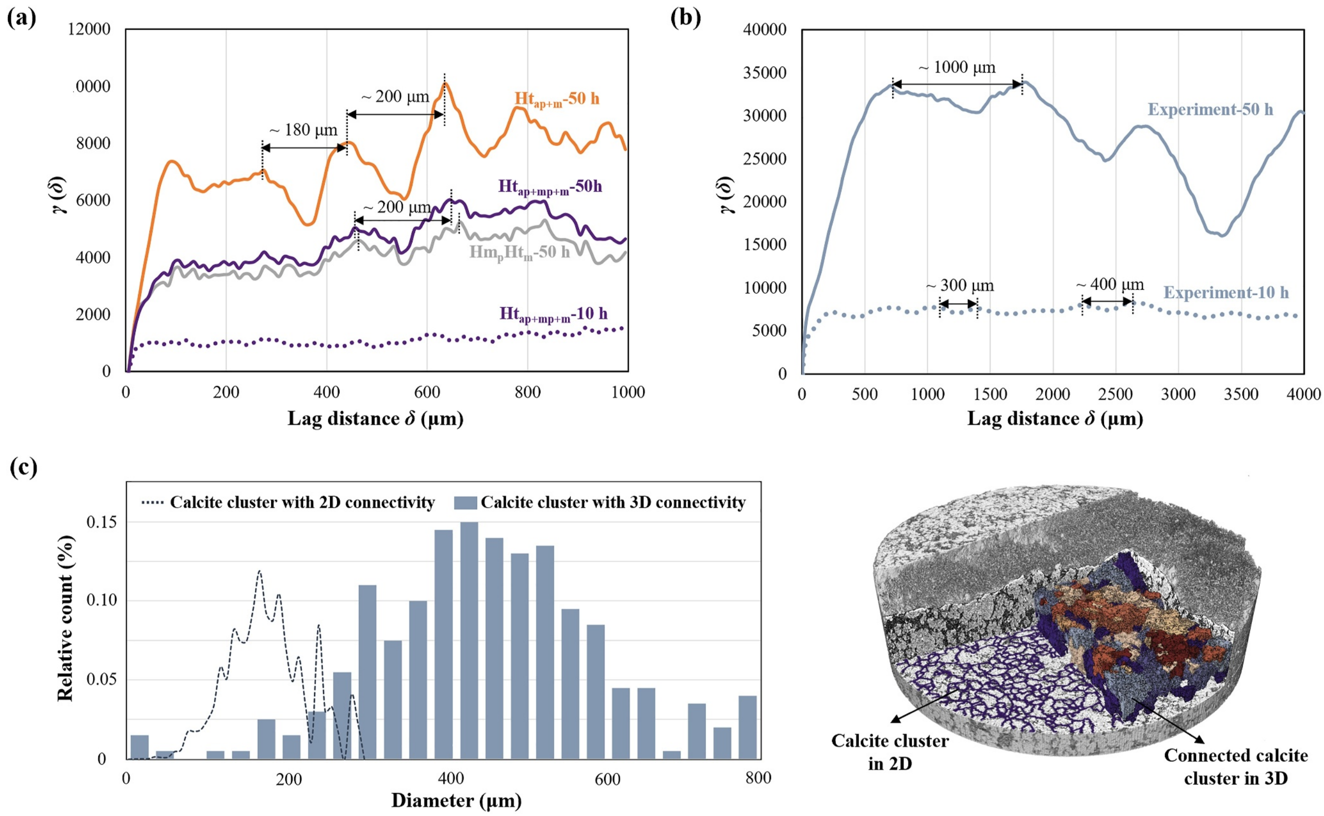

Figure 5. Semi-variograms of the AL thickness analyzed from (a) different simulation scenarios and (b) the experiment, and (c) size distributions of calcite clusters in terms of 2D and 3D connectivity. The black arrows in (a) and (b) highlight the periodic feature of semi-variograms. In panel (c), the connected calcite clusters shown with the same color are identified using the method of watershed separation (more details in Supporting Information S1).

3.2.2. Contribution of Mineralogical Heterogeneity

Compared to the pore structure effects, the heterogeneity in mineralogy plays a dominant role in spatial variations in AL. As shown in Figure 4, the "channeling" structure of calcite-depleted regions and the jaggedness of reactive fronts can only be observed in scenarios when considering a non-uniform distribution of calcite minerals. Additionally, the variations in the AL profiles from simulations of all four slices and profiles from the transmission data highlights a close dependence on the initial rock texture (Figure S5 in Supporting Information S1). To better illustrate how mineralogical heterogeneity controls spatial variations of AL, we examined the semi-variograms of AL thickness in different simulation scenarios and the experiment. The sill (σ^2), the value of semi-variance when it reaches a plateau, provides an indicator of the overall spatial variability of AL thickness; and the corresponding lag distance, aka the range (φ), represents the spatial correlation length (Noiriel et al., 2013). Information about the determination of semi-variograms and related parameters (i.e., the sill and the range) is provided in Text S6 of the Supporting Information S1 (Mert & Dag, 2017).

For all simulations with mineralogical spatial heterogeneity, σ^2 and φ increase over time, indicating an increase of autocorrelation and spatial variation (Figure 5); whereas for the $Ht_{ap+mp}Hm_m$ and Hm_{p+m} scenarios, a plateau did not develop in the semi-variogram because of the uniform AL thickness, and thus they are not discussed further. The increase in σ^2 and φ is consistent with the widening of the channeling features in the experimental maps of AL thickness (shown in Figure S1b of the Supporting Information S1) and the trends in the semi-variogram of the experimental data. The absolute values of σ^2 and φ are larger in the experiment as expected, because the dissolution patterns in the experiment reflect the superimposition of additional spatial heterogeneities in the third dimension.

One important feature revealed by the cyclicity in the "plateau" portion of the semi-variogram is the presence of periodic variations in AL thickness over distance (highlighted by the black arrows in Figures 5a and 5b). Such

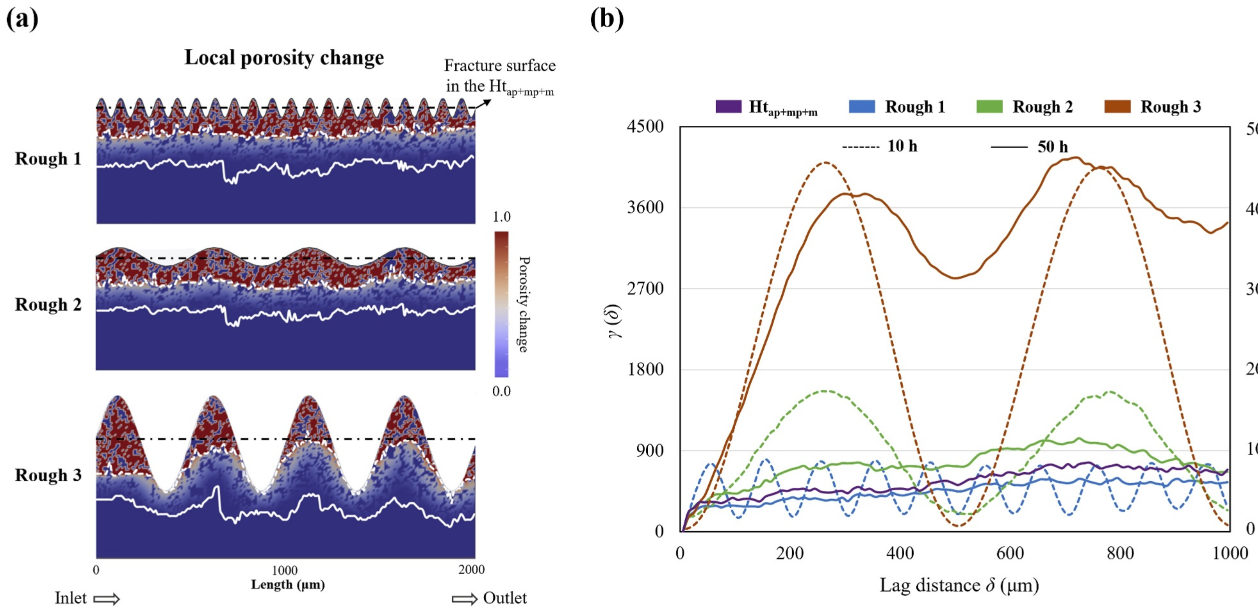

Figure 6. (a) Maps of porosity increase at 50 hr and semi-variograms of the reaction fronts for different rough fracture scenarios. The white dashed lines and white solid lines in (a) show the positions of the maximum reaction rate (the reaction front) and a low reaction rate (1×10^{-8} mol/m^2/s) (the diffusion front), respectively. The black dash-dotted lines display the initial fracture surface in the $Ht_{ap+mp+m}$ scenario.

sinusoidal features are indicators of fine-scale variability (David, 1977; Journel & Huijbregts, 1978). The cyclic wavelength of the modeling data is 150–200 μm (Figure 5a), which is comparable to the size of calcite clusters identified from XCT images. Figure 5c shows that calcite cluster size determined from the 2D slices have a distribution between 50 and 250 μm and a peak of 130 μm. Considering the 3D connectivity, more touching calcite clusters can be grouped into a “watershed” (see Text S7 in Supporting Information S1, for more details). Consequently, the size of connected calcite clusters grows and the average value is 560 μm. A larger wavelength of 800–1,000 μm was observed in the experimental AL thicknesses (Figure 5b), supporting a larger autocorrelation resulting from the increased connectivity of reactive minerals. Given that the simulations are 2D and thus connections in the third dimension are missing, the periodic feature of AL thickness confirms semi-quantitatively its strong correlation with the textural characteristics of the reactive mineral.

We further investigated the impacts of matrix heterogeneity in rough fractures. Fracture surface roughness was generated using a single sine wave with a wavelength of 100 or 500 μm, comparable to the size of the calcite cluster size, and an amplitude of 50 or 250 μm. The simulations are referred to Rough 1–3, and more details of the simulation setup are available in Supporting Information S1 (Text S5). The results of local porosity change at 50 hr for different scenarios show that fracture surface roughness results in larger variations in local AL thickness that is constrained by the fracture surface itself (Figure 6a). However, the difference in average AL thickness (at 50 hr) is up to 4.8%, whereas the difference is up to 26.8% with varied degrees of matrix heterogeneity. This is different from previous studies of impervious rough fractures, which showed that surface roughness increases surface area and thus the overall reaction (Deng, Molins, et al., 2018; Deng, Steefel, et al., 2018). Moreover, the impact of surface roughness on reactions is increasingly limited as the reaction front migrates downward into the matrix. This is evident in the shape of the reaction front and the weakening of the cyclic features in the semi-variograms of the reaction fronts over time (Figure 6b).

In summary, a better connectivity of calcite minerals, that is, more touching clusters with a limited spatial heterogeneity, can significantly improve the rate and the extent of reaction and thus AL development. The texture of reactive minerals, like rough fracture surface, can play a critical role in spatial variations of AL, and support the hypothesis of the experimental study that the spatial distribution of calcite clusters triggers instabilities in reaction fronts, leading to the channelization of the reactive front below the flat fracture. Such channeling contributes to a relatively larger degree of permeability increase and a lower bulk reaction rate.

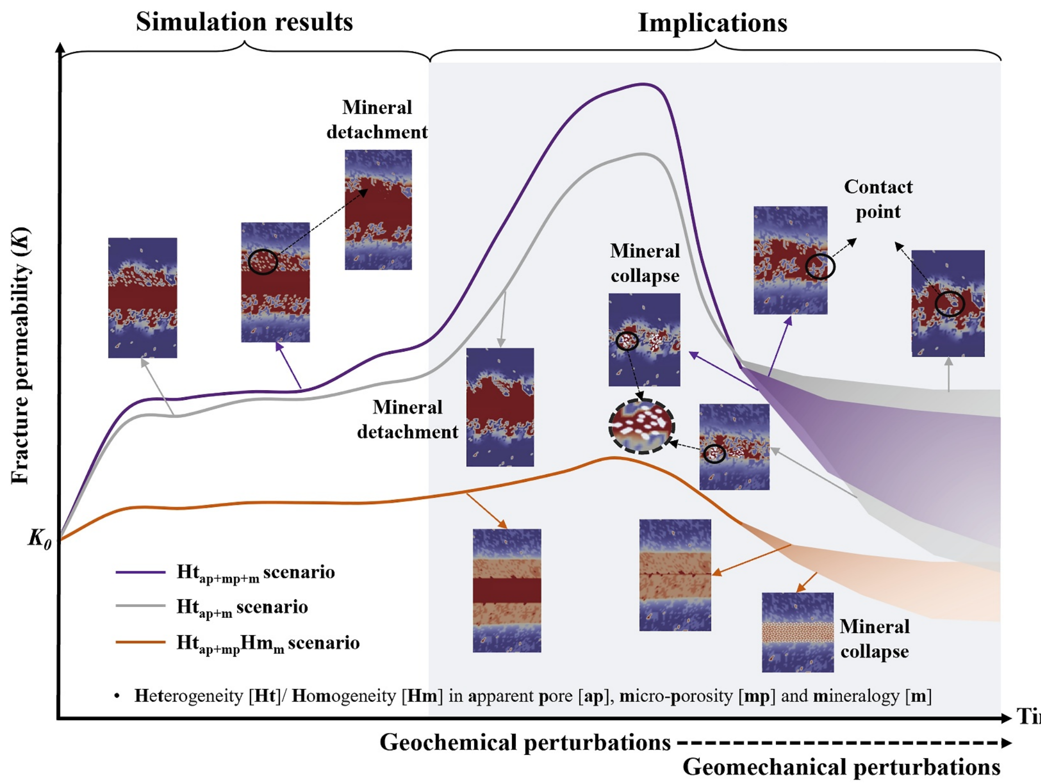

Figure 7. Schematic diagram of fracture permeability evolution regarding matrix heterogeneity during the coupled geochemical and geomechanical processes.

3.3. Implications for the Coupling of Geochemical and Geomechanical Perturbations

Our simulation results highlight that the degree of matrix heterogeneity can lead to varied patterns of fractured media alteration, and such differences can propagate and potentially be amplified during a long-term reaction or in the presence of mechanical perturbations. Figure 7 illustrates different trajectories of fracture permeability evolution that can occur due to matrix heterogeneity. Permeability enhancement is expected to grow when the AL detach (Khan et al., 2022). Erosion of an evolving AL, although was not observed in the microfluidic experiment used for our simulations, is a well-established mechanism for mineral removal and can result in the enlargement of fracture aperture and continued AL development. Previous studies have confirmed that matrix disaggregation can be triggered when the fast-reacting mineral exceeds a threshold (e.g., 35%) or distributes nonuniformly (Deng et al., 2017; Fang et al., 2018). In addition, a thicker AL, which can result from the presence of a larger micro-porosity, may also facilitate the erosion. In contrast, more uniform dissolution in the ALs leads to less fluctuations in permeability.

When geomechanical perturbations are involved, the fracture aperture can switch from opening to closing, reducing permeability. In general, fracture closure induced by normal stress can be deterred due to the persistence of asperities, and an increase in cohesive interlocking asperities can lead to higher frictional strength and stability during shear as well (Fang et al., 2018). In presence of AL development, if islands of thin AL spots or AL-free spots are left behind, resistance against fracture closure remain. These behaviors can be more evident with more AL asperities, for example, resulting from the absence of micro-porosity, or increased surface roughness of initial fractures. If a uniform layer of AL develops (i.e., in the Hm_{p+m} scenario), the entire AL may collapse under stress and the remaining minerals may create a layer of fine particles filling the fracture aperture, reducing fracture permeability and frictional strength (Spokas et al., 2019).

Therefore, inaccurate predictions of AL development, including the spatial features of thickness and structure, can lead to misinformation of fractured media evolution, especially when the coupling of chemical and mechanical processes.

4. Conclusions

Rock heterogeneities, that is, the abundance and spatial variations in pore space and mineralogy, exist in natural subsurface systems and complicate geochemically driven evolution of fractured media. This study, enabled by a well-characterized experimental data set and a micro-continuum reactive transport model, systematically investigated the respective contribution of structural and mineralogical heterogeneity on the development of AL at fracture-matrix interfaces and the resulting changes in bulk properties of the fractured media. The findings from this work highlight that the size and connectivity of fast-reacting mineral clusters mainly control spatial variations of AL, whereas the presence of micro-porosity can significantly improve the accessibility of reactive minerals, and thus thickening of the AL.

These insights bear important implications. First, they imply that textural information of matrix can be used as a priori data for the predictions of geochemical alteration of fracture-matrix interfaces and spatio-temporal features of the AL development. It has been increasingly recognized that rocks with minerals of differing reactivities but comparable abundance can trigger multiple mechanisms of fracture alteration that control the evolution of chemical-physical properties (Al-Khulaifi et al., 2019; Deng, Molins, et al., 2018; Deng, Steefel, et al., 2018). Such impacts of matrix properties can be more evident in a long-term evolution regardless of fracture surface roughness. Therefore, numerical investigation at small scale can shed light on how matrix heterogeneity gives rise to the difference in the rates between laboratory and field scales and changes the permeability-porosity relationship.

Second, in cases where geochemically triggered mechanical alteration are expected, characterization and simulations that resolve mineralogical features at a resolution that is finer than the characteristic length of the textures and thus the resulting spatial variations in AL will be required. A higher degree of AL spatial variations may facilitate the disaggregation of remaining minerals and thus enlarge the fracture aperture, or suppress the fracture closure with more bridging asperities. In this regard, the internal structure of developed ALs has a profound impact on long-term alterations in fractured media, especially when mechanical perturbations coexist.

Data Availability Statement

Input files and simulation results for the reactive transport modeling of the fractured media are available in Zhang et al. (2024).

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